Solving analytical queries on Redshift Cluster

Here, you have to write the query used for solving the question and the screenshots of the table which is outputted after the query is run on the AWS Redshift Query editor UI.

1. **Top 10 ATMs where most transactions are in the 'inactive' state**

```
SELECT
   a.atm_number,
   a.atm_manufacturer,
   l.location,
   COUNT(*) AS total_transaction_count,
   COUNT(*) AS inactive_txn_count
FROM fact_atm_trans f
LEFT JOIN dim_atm a ON f.atm_id = a.atm_id
LEFT JOIN dim_location l ON a.atm_location_id = l.location_id
WHERE f.atm_status = 'Inactive'
GROUP BY a.atm_number, a.atm_manufacturer, l.location
ORDER BY inactive_txn_count DESC
LIMIT 10;
```

atm_number	atm_manufacturer	location	total_transaction_co...	inactive_txn_count
16	NCR	Skive	44043	44043
12	NCR	ÃƒËœsterÃƒÂ¥ Duus	33982	33982
2	NCR	Vejgaard	33725	33725
88	NCR	Storcenter indg. A	32183	32183
30	NCR	NykÃƒÂ¸bing Mors	30883	30883
52	NCR	FarsÃƒÂ¸	27361	27361
50	NCR	Aarhus	23416	23416
29	NCR	Skelagervej 15	20773	20773
81	NCR	Spar KÃƒÂ¸bmand Tornh...	20148	20148
102	NCR	Aalborg Storcenter Afd	18297	18297

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2. **Number of ATM failures corresponding to the different weather conditions recorded at the time of the transactions**

```
SELECT
   f.weather_main,
   COUNT(*) AS total_transaction_count,
   SUM(
      CASE
         WHEN f.atm_status = 'Inactive' THEN 1
         ELSE 0
      END
   ) AS inactive_count,
   ROUND(
      SUM(
         CASE
            WHEN f.atm_status = 'Inactive' THEN 1
            ELSE 0
         END
      ) * 100.0 / COUNT(*),
      4
   ) AS inactive_count_percent
FROM fact_atm_trans f
WHERE weather_main IS NOT NULL
GROUP BY f.weather_main
ORDER BY inactive_count_percent DESC;
```

weather_main	total_transaction_co...	inactive_count	inactive_count_percent
Snow	23405	4813	20.564
Fog	18174	3729	20.5183
Clouds	1181901	194027	16.4165
Rain	545135	86017	15.779
Clear	543949	85531	15.7241
Mist	82801	12864	15.536
Thunderstorm	2549	361	14.1624
Drizzle	62530	8670	13.8653
TORNADO	38	1	2.6316
Haze	3	0	0

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3. **Top 10 ATMs with the most number of transactions throughout the year**

```
SELECT
   a.atm_number,
   a.atm_manufacturer,
   l.location,
   COUNT(*) AS total_transaction_count
FROM fact_atm_trans f
JOIN dim_atm a
   ON f.atm_id = a.atm_id
JOIN dim_location l
   ON a.atm_location_id = l.location_id
GROUP BY
   a.atm_number,
   a.atm_manufacturer,
   l.location
ORDER BY total_transaction_count DESC
LIMIT 10;
```

atm_number	atm_manufacturer	location	total_transaction_co...
39	NCR	Svenstrup	55380
20	NCR	Bispensgade	54211
10	NCR	NÃƒÂ¸rresundby	53794
24	NCR	Hobro	53378
45	NCR	Abildgaard	53198
16	NCR	Skive	44043
40	Diebold Nixdorf	Frederikshavn	43767
1	NCR	NÃƒÂ¦stved	42787
41	Diebold Nixdorf	Skagen	42732
48	Diebold Nixdorf	BrÃƒÂ¸nderslev	42493

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4. **Number of overall ATM transactions going inactive per month for each month**

```
SELECT
   d.year,
   d.month,
   COUNT(*) AS total_transaction_count,
   SUM(
      CASE
         WHEN f.atm_status = 'Inactive' THEN 1
         ELSE 0
      END
   ) AS inactive_count,
   ROUND(
      SUM(
         CASE
            WHEN f.atm_status = 'Inactive' THEN 1
            ELSE 0
         END
      ) * 100.0 / COUNT(*),
      4
   ) AS inactive_count_percent
FROM fact_atm_trans f
LEFT JOIN dim_date d
   ON f.date_id = d.date_id
GROUP BY
   d.year,
   d.month
ORDER BY
   d.year,
   d.month;
```

year	month	total_transaction_co...	inactive_count	inactive_count_percent
2017	April	218865	41830	19.1122
2017	August	217218	36713	16.9015
2017	December	197048	20476	10.3914
2017	February	182659	36656	20.068
2017	January	180195	35953	19.9523
2017	July	227682	38139	16.751
2017	June	225166	36789	16.3386
2017	March	209586	41046	19.5843
2017	May	222418	37679	16.9406
2017	November	193967	21684	11.1792

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5. **Top 10 ATMs with the highest total withdrawn amount throughout the year**

```
SELECT
   a.atm_number,
   a.atm_manufacturer,
   l.location,
   SUM(f.transaction_amount) AS total_transaction_amount
FROM fact_atm_trans f
JOIN dim_atm a
   ON f.atm_id = a.atm_id
JOIN dim_location l
   ON a.atm_location_id = l.location_id
GROUP BY
   a.atm_number,
   a.atm_manufacturer,
   l.location
ORDER BY
   total_transaction_amount DESC
LIMIT 10;
```

atm_number	atm_manufacturer	location	total_transaction_am...
39	NCR	Svenstrup	277097637
20	NCR	Bispensgade	271008803
24	NCR	Hobro	268289882
10	NCR	NÃƒÂ¸rresundby	267379103
45	NCR	Abildgaard	265639616
16	NCR	Skive	220677013
40	Diebold Nixdorf	Frederikshavn	219812287
41	Diebold Nixdorf	Skagen	214127315
1	NCR	NÃƒÂ¦stved	213721117
48	Diebold Nixdorf	BrÃƒÂ¸nderslev	212883099

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6. **Number of failed ATM transactions across various card types**

```
SELECT
   c.card_type,
   COUNT(*) AS total_transaction_count,
   SUM(
      CASE
         WHEN f.atm_status = 'Inactive' THEN 1
         ELSE 0
      END
   ) AS inactive_count,
   ROUND(
      SUM(
         CASE
            WHEN f.atm_status = 'Inactive' THEN 1
            ELSE 0
         END
      ) * 100.0 / COUNT(*),
      4
   ) AS inactive_count_percent
FROM fact_atm_trans f
JOIN dim_card_type c
   ON f.card_type_id = c.card_type_id
GROUP BY
   c.card_type
ORDER BY
   inactive_count_percent DESC;
```

card_type	total_transaction_co...	inactive_count	inactive_count_percent
Mastercard - on-us	458226	86000	18.768
VISA	170828	30713	17.9789
Dankort - on-us	143813	24680	17.1612
CIRRUS	17362	2953	17.0084
HÃƒÂ¦vekort - on-us	62487	10331	16.533
Dankort	28581	4557	15.9442
MasterCard	400507	63482	15.8504
Visa Dankort - on-us	748805	112972	15.087
HÃƒÂ¦vekort	8459	1208	14.2806
Visa Dankort	427840	60547	14.1518

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7. **Number of transactions happening on an ATM on weekdays and on weekends throughout the year. Order this by the ATM_number, ATM_manufacturer, location, weekend_flag and then total_transaction_count**

```
SELECT
   a.atm_number,
   a.atm_manufacturer,
   l.location,
   CASE
      WHEN d.weekday IN ('Saturday', 'Sunday') THEN 'Weekend'
      ELSE 'Weekday'
   END AS weekend_flag,
   COUNT(*) AS total_transaction_count
FROM fact_atm_trans f
JOIN dim_atm a
   ON f.atm_id = a.atm_id
JOIN dim_location l
   ON a.atm_location_id = l.location_id
JOIN dim_date d
   ON f.date_id = d.date_id
GROUP BY
   a.atm_number,
   a.atm_manufacturer,
   l.location,
   CASE
      WHEN d.weekday IN ('Saturday', 'Sunday') THEN 'Weekend'
      ELSE 'Weekday'
   END
ORDER BY
   a.atm_number,
   a.atm_manufacturer,
   l.location,
   weekend_flag,
   total_transaction_count DESC;
```

atm_number	atm_manufacturer	location	weekend_flag	total_transaction_co...
1	NCR	NÃƒÂ¦stved	Weekday	32711
1	NCR	NÃƒÂ¦stved	Weekend	10076
10	NCR	NÃƒÂ¸rresundby	Weekday	41667
10	NCR	NÃƒÂ¸rresundby	Weekend	12127
100	NCR	Intern Skive	Weekday	17812
100	NCR	Intern Skive	Weekend	1
101	NCR	Bryggen Vejle	Weekday	11693
101	NCR	Bryggen Vejle	Weekend	3247
102	NCR	Aalborg Storcenter Afd	Weekday	14556
102	NCR	Aalborg Storcenter Afd	Weekend	3741

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8. **Most active day in each ATMs from location "Vejgaard"**

```
WITH weekday_counts AS (
   SELECT
      a.atm_number,
      a.atm_manufacturer,
      l.location,
      d.weekday,
      COUNT(*) AS total_transaction_count
   FROM fact_atm_trans f
   JOIN dim_atm a
      ON f.atm_id = a.atm_id
   JOIN dim_location l
      ON a.atm_location_id = l.location_id
   JOIN dim_date d
      ON f.date_id = d.date_id
   WHERE l.location = 'Vejgaard'
   GROUP BY
      a.atm_number,
      a.atm_manufacturer,
      l.location,
      d.weekday
),
ranked AS (
   SELECT *,
         ROW_NUMBER() OVER (
            PARTITION BY atm_number
            ORDER BY total_transaction_count DESC
         ) AS rn
   FROM weekday_counts
)
SELECT
   atm_number,
   atm_manufacturer,
   location,
   weekday,
   total_transaction_count
FROM ranked
WHERE rn = 1
ORDER BY atm_number;
```

atm_number	atm_manufacturer	location	weekday	total_transaction_co...
103	Diebold Nixdorf	Vejgaard	Friday	4757
2	NCR	Vejgaard	Friday	6290

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